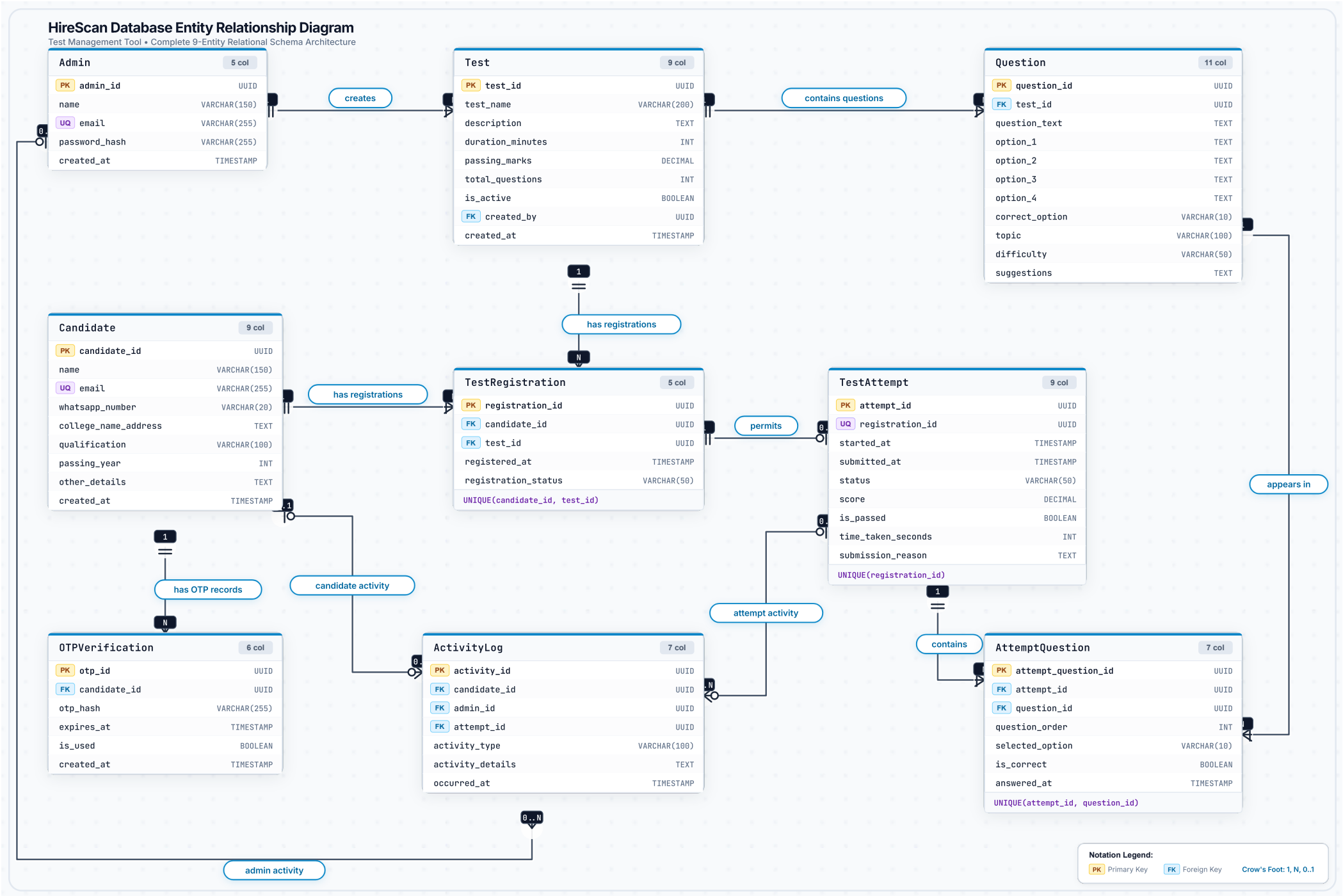

HireScan Database Entity Relationship Diagram
Test Management Tool • Complete 9-Entity Relational Schema Architecture
Admin 5 col
PK admin_id UUID
name VARCHAR(150)
UQ email VARCHAR(255)
password_hash VARCHAR(255)
created_at TIMESTAMP
creates
Test 9 col
PK test_id UUID
test_name VARCHAR(200)
description TEXT
duration_minutes INT
passing_marks DECIMAL
total_questions INT
is_active BOOLEAN
FK created_by UUID
created_at TIMESTAMP
contains questions
Question 11 col
PK question_id UUID
FK test_id UUID
question_text TEXT
option_1 TEXT
option_2 TEXT
option_3 TEXT
option_4 TEXT
correct_option VARCHAR(10)
topic VARCHAR(100)
difficulty VARCHAR(50)
suggestions TEXT
has registrations
Candidate 9 col
PK candidate_id UUID
name VARCHAR(150)
UQ email VARCHAR(255)
whatsapp_number VARCHAR(20)
college_name_address TEXT
qualification VARCHAR(100)
passing_year INT
other_details TEXT
created_at TIMESTAMP
has registrations
TestRegistration 5 col
PK registration_id UUID
FK candidate_id UUID
FK test_id UUID
registered_at TIMESTAMP
registration_status VARCHAR(50)
UNIQUE(candidate_id, test_id)
permits
TestAttempt 9 col
PK attempt_id UUID
UQ registration_id UUID
started_at TIMESTAMP
submitted_at TIMESTAMP
status VARCHAR(50)
score DECIMAL
is_passed BOOLEAN
time_taken_seconds INT
submission_reason TEXT
UNIQUE(registration_id)
appears in
has OTP records
OTPVerification 6 col
PK otp_id UUID
FK candidate_id UUID
otp_hash VARCHAR(255)
expires_at TIMESTAMP
is_used BOOLEAN
created_at TIMESTAMP
candidate activity
attempt activity
ActivityLog 7 col
PK activity_id UUID
FK candidate_id UUID
FK admin_id UUID
FK attempt_id UUID
activity_type VARCHAR(100)
activity_details TEXT
occurred_at TIMESTAMP
contains
AttemptQuestion 7 col
PK attempt_question_id UUID
FK attempt_id UUID
FK question_id UUID
question_order INT
selected_option VARCHAR(10)
is_correct BOOLEAN
answered_at TIMESTAMP
UNIQUE(attempt_id, question_id)
admin activity
Notation Legend:
PK Primary Key
FK Foreign Key
Crow's Foot: 1, N, 0..1

Entity Specifications & Attribute Dictionary

9 Tables • Normalized Relational Architecture

Admin

System administrators and evaluators

5 attributes

PK	admin_id	UUID
	name	VARCHAR(150)
UQ	email	VARCHAR(255)
	password_hash	VARCHAR(255)
	created_at	TIMESTAMP

Test

Assessment blueprints and configurations

9 attributes

PK	test_id	UUID
	test_name	VARCHAR(200)
	description	TEXT
	duration_minutes	INT
	passing_marks	DECIMAL
	total_questions	INT
	is_active	BOOLEAN
FK	created_by	UUID
	created_at	TIMESTAMP

Question

Question repository for assessments

11 attributes

PK	question_id	UUID
FK	test_id	UUID
	question_text	TEXT
	option_1	TEXT
	option_2	TEXT
	option_3	TEXT
	option_4	TEXT
	correct_option	VARCHAR(10)
	topic	VARCHAR(100)
	difficulty	VARCHAR(50)
	suggestions	TEXT

Candidate

Job applicants and examinees

9 attributes

PK	candidate_id	UUID

TestRegistration

Candidate enrolment in tests

5 attributes

PK	registration_id	UUID

TestAttempt

Assessment sitting session (1 per registration)

9 attributes

PK	attempt_id	UUID
FK UQ	registration_id	UUID
	started_at	TIMESTAMP

	name	VARCHAR(150)
UQ	email	VARCHAR(255)
	whatsapp_number	VARCHAR(20)
	college_name_address	TEXT
	qualification	VARCHAR(100)
	passing_year	INT
	other_details	TEXT
	created_at	TIMESTAMP

FK	candidate_id	UUID
FK	test_id	UUID
	registered_at	TIMESTAMP
	registration_status	VARCHAR(50)

UNIQUE(candidate_id, test_id)

	submitted_at	TIMESTAMP
	status	VARCHAR(50)
	score	DECIMAL
	is_passed	BOOLEAN
	time_taken_seconds	INT
	submission_reason	TEXT

UNIQUE(registration_id)

AttemptQuestion

Response per question in attempt

7 attributes

PK	**attempt_question_id**	UUID
FK	attempt_id	UUID
FK	question_id	UUID
	question_order	INT
	selected_option	VARCHAR(10)
	is_correct	BOOLEAN
	answered_at	TIMESTAMP

UNIQUE(attempt_id, question_id)

OTPVerification

Authentication one-time passwords

6 attributes

PK	**otp_id**	UUID
FK	candidate_id	UUID
	otp_hash	VARCHAR(255)
	expires_at	TIMESTAMP
	is_used	BOOLEAN
	created_at	TIMESTAMP

ActivityLog

Audit log with independently nullable actors

7 attributes

PK	**activity_id**	UUID
FK	candidate_id	UUID
FK	admin_id	UUID
FK	attempt_id	UUID
	activity_type	VARCHAR(100)
	activity_details	TEXT
	occurred_at	TIMESTAMP